AI 454 - NLP ASSIGNMENT 1



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Ahmad Hazaimeh ID:162104

Zeid Al-Suleiman ID:166203

Table 1: Semantic Analogy Test Results (300-dimensional models):

Test #	Test	Expected Answer	GloVe 300 Result	Word2Vec 300 Result	FastText 300 Result
1	france : paris :: italy : ?	rome	rome	OOV	rome
2	germany : berlin :: spain : ?	madrid	madrid	OOV	barcelona
3	japan : tokyo :: china : ?	beijing	beijing	OOV	shanghai
4	egypt : cairo :: greece : ?	athens	athens	OOV	gtk
5	india : new delhi :: pakistan : ?	islamabad	OOV	OOV	OOV
6	brazil : brasilia :: argentina : ?	buenos aires	buenos	OOV	OOV
7	russia : moscow :: poland : ?	warsaw	warsaw	OOV	prague
8	canada : ottawa :: australia : ?	canberra	canberra	OOV	melbourne
9	turkey : ankara :: iran : ?	tehran	tehran	OOV	OOV
10	netherlands : amsterdam :: portugal : ?	lisbon	lisbon	OOV	barcelona

Table 2: Syntactic Analogy Test Results (300-dimensional models)

Test #	Test	Expected Answer	GloVe 300 Result	Word2Vec 300 Result	FastText 300 Result
1	kind : kinder :: brave : ?	braver	gentler	braver	braver
2	smart : smarter :: tall : ?	taller	taller	taller	taller
3	short : shorter :: strong : ?	stronger	stronger	stronger	stronger
4	weak : weaker :: quiet : ?	quieter	quieter	quieter	quieter
5	messy : messier :: calm : ?	calmer	gretzky	calmer	calmer
6	wise : wiser :: cold : ?	colder	colder	colder	colder
7	noisy : noisier :: clumsy : ?	clumsier	sillier	cleverer	ham-fisted

8	lazy : lazier :: bossy : ?	bossier	scheckter	OOV	meaner
9	young : younger :: old : ?	older	older	older	older
10	hot : hotter :: small : ?	smaller	smaller	smaller	smaller

Table 3: Analogy Top-1 Accuracy (%)

Analogy Type	GloVe 300 Accuracy	Word2Vec 300 Accuracy	FastText 300 Accuracy
Semantic	80.00%	0.00%	10.00%
Syntactic	60.00%	80.00%	80.00%

Table 4: Analogy Top-5 Accuracy (%)

Analogy Type	GloVe 300 Accuracy	Word2Vec 300 Accuracy	FastText 300 Accuracy
Semantic	80.00%	0.00%	40.00%
Syntactic	60.00%	80.00%	80.00%

Part 2: Intrinsic Evaluation of Word Embeddings: GloVe (50 vs 100 vs 300)

Table 5: GloVe Dimension Comparison (Semantic Analogies):

Test #	Test	Expected Answer	GloVe 50 Result	GloVe 100 Result	GloVe 300 Result
1	france : paris :: italy : ?	rome	rome	rome	rome
2	germany : berlin :: spain : ?	madrid	rome	madrid	madrid
3	japan : tokyo :: china : ?	beijing	beijing	beijing	beijing
4	egypt : cairo :: greece : ?	athens	athens	athens	athens
5	india : new delhi :: pakistan : ?	islamabad	OOV	OOV	OOV

6	brazil : brasilia :: argentina : ?	buenos aires	lima	buenos	buenos
7	russia : moscow :: poland : ?	warsaw	warsaw	warsaw	warsaw
8	canada : ottawa :: australia : ?	canberra	brisbane	brisbane	canberra
9	turkey : ankara :: iran : ?	tehran	tehran	tehran	tehran
10	netherlands : amsterdam :: portugal : ?	lisbon	lisbon	lisbon	lisbon

Table 6: GloVe Dimension Comparison (Syntactic Analogies):

Test #	Test	Expected Answer	GloVe 50 Result	GloVe 100 Result	GloVe 300 Result
1	kind : kinder :: brave : ?	braver	przewalskii	schwalbe	gentler
2	smart : smarter :: tall : ?	taller	taller	taller	taller
3	short : shorter :: strong : ?	stronger	stronger	stronger	stronger
4	weak : weaker :: quiet : ?	quieter	calm	subdued	quieter
5	messy : messier :: calm : ?	calmer	moussa	gretzky	gretzky

6	wise : wiser :: cold : ?	colder	hotter	cooler	colder
7	noisy : noisier :: clumsy : ?	clumsier	horrendously	sillier	sillier
8	lazy : lazier :: bossy : ?	bossier	konchellah	skaife	scheckter
9	young : younger :: old : ?	older	father	older	older
10	hot : hotter :: small : ?	smaller	larger	smaller	smaller

Table 7: GloVe Top-1 Accuracy:

Analogy Type	GloVe 50 Accuracy	GloVe 100 Accuracy	GloVe 300 Accuracy
Semantic	60.00%	70.00%	80.00%
Syntactic	20.00%	40.00%	60.00%

Table 8: GloVe Top-5 Accuracy:

Analogy Type	GloVe 50 Accuracy	GloVe 100 Accuracy	GloVe 300 Accuracy
Semantic	70.00%	80.00%	80.00%
Syntactic	40.00%	50.00%	60.00%

Part 3: Visualization of Word Embeddings using GloVe 100 vs Word2Vec 100

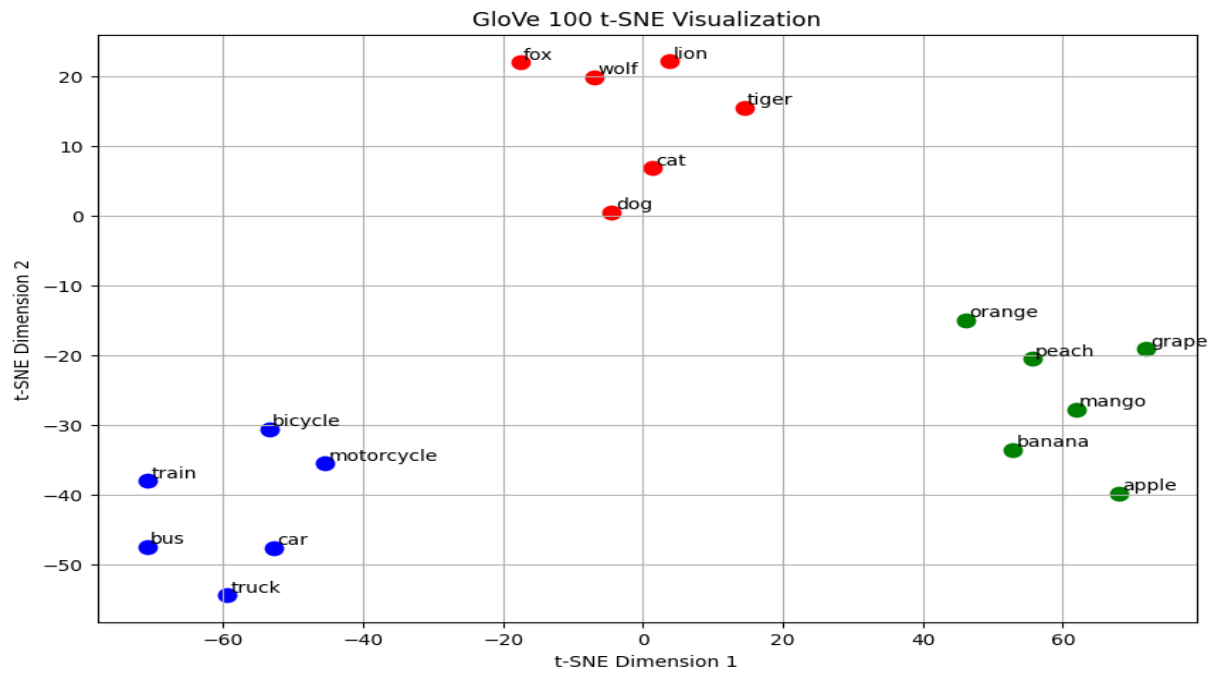
1. Design of Word Sets:

- **Animals**: {cat, dog, lion, tiger, wolf, fox}.
- **Fruits**: {apple, banana, orange, grape, mango, peach}.
- **Vehicles**: {car, bus, train, truck, bicycle, motorcycle}.

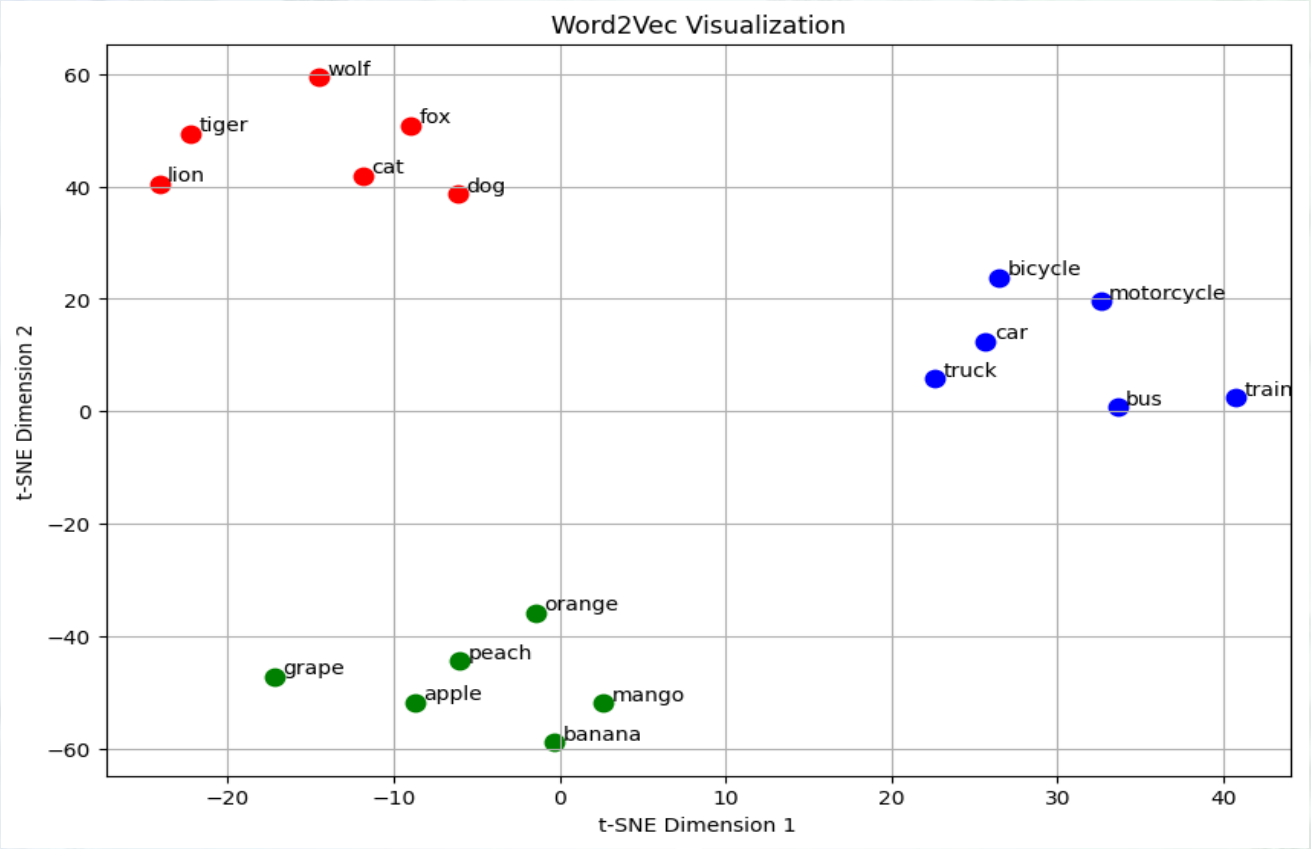
2. Methodology:

- Obtained 100d embeddings for all words from **GloVe** and **Word2Vec**.
- Applied **t-SNE** to reduce dimensions to 2D for visualization.
- Labeled each word and distinguished sets using specific colors.

GloVe Visualization:



Word2Vec Visualization:



3. Analysis & Comparison:

- **Clustering**: In both models, related words clustered effectively, and the three sets were well-separated with no overlap.
- **Model Comparison**: **Word2Vec** produced slightly better clustering.
- **Justification**: Word2Vec showed tighter, more compact clusters and larger distances between different categories compared to GloVe.

Final Conclusion:

This assignment evaluated **Word2Vec**, **GloVe**, and **FastText** through analogy tasks and t-SNE visualization. Our results show that **GloVe** excels in semantic tasks, while **Word2Vec** and **FastText** are superior for syntactic patterns. Increasing dimensionality from 50 to 300 improved accuracy but raised computational costs. Visualization confirmed that embeddings capture semantic similarity, with **Word2Vec** producing the most compact clusters. Ultimately, model selection depends on the specific trade-off between accuracy and efficiency.